

Video-of-Thought: Step-by-Step Video Reasoning from Perception to Cognition

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01 Reasoning

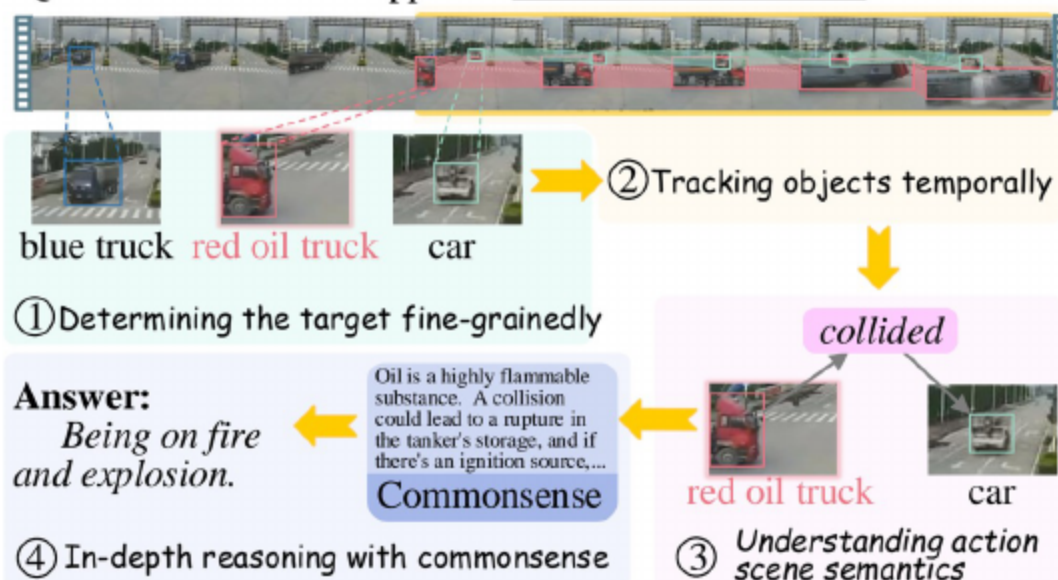
QUESTION: What will happen to the red oil tanker truck?



- **Gap:** models excel at simple perception but struggle with complex video cognition
- Complex reasoning needs spatiotemporal understanding plus implied semantics
- Real-world applications demand reasoning over interactions, causality, and intent
- Shallow recognition is insufficient; deeper scene semantics and commonsense are required
- **Challenge:** bridging pixel-level cues with high-level narrative understanding

02 Human-like Multi-step Reasoning: Core Idea

QUESTION: What will happen to the red oil tanker truck?



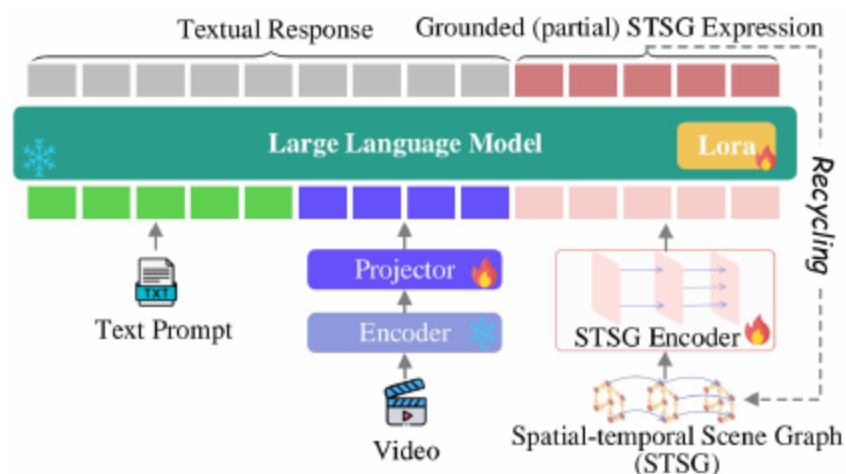
- Humans: identify → track and ground → analyze interactions → integrate commonsense
- Progression from pixel grounding to cognitive semantics with interpretable steps
- **Outcome:** coherent, verified answers with explicit rationales
- **Benefit:** reduces ambiguity by decomposing complex queries
- Targets, trajectories, and relations form the backbone of reasoning

03 CoT for Video MLLMs is Under-explored



- MLLMs advanced; video MLLMs emerging
- Classic CoT helps text tasks; video-specific CoT scarce
- Need a video-native CoT that decomposes problems with temporal grounding
- **Limitation:** naive step-by-step prompts underperform for video
- **Opportunity:** combine structure from scene graphs with LLM commonsense

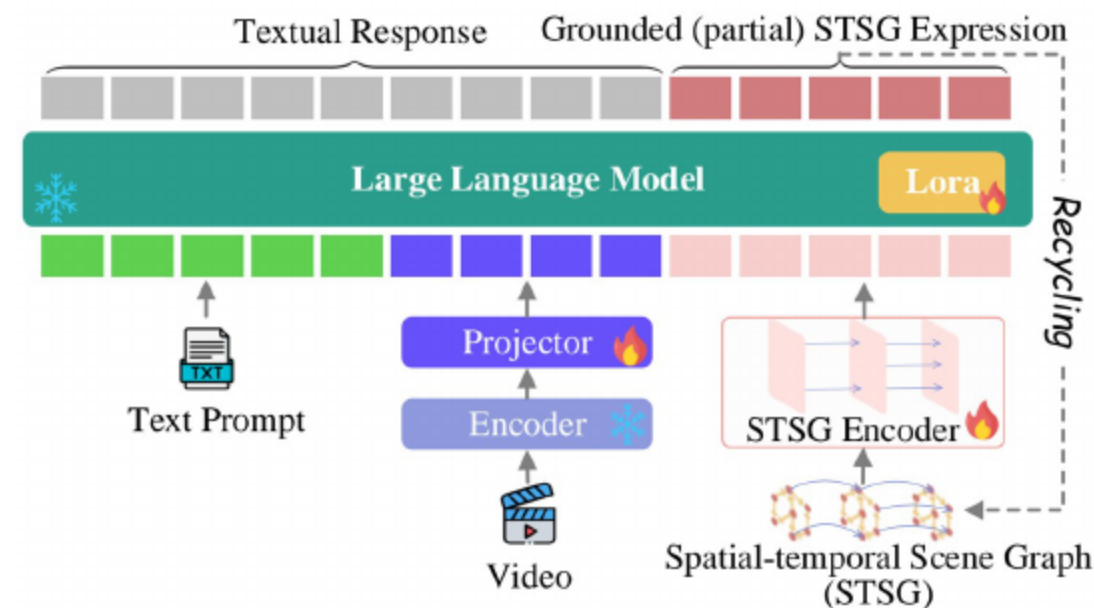
04 MotionEpic + Video-of-Thought (VoT)



- **MotionEpic**: video MLLM that understands and generates STSGs
- **VoT**: chains video reasoning from grounding to cognition
- Grounding-aware tuning enables STSG-free downstream inference
- **Gain**: interpretable rationales through generated structure
- Unified pipeline for perception, interaction analysis, and answer verification

[Ji et al., 2020]

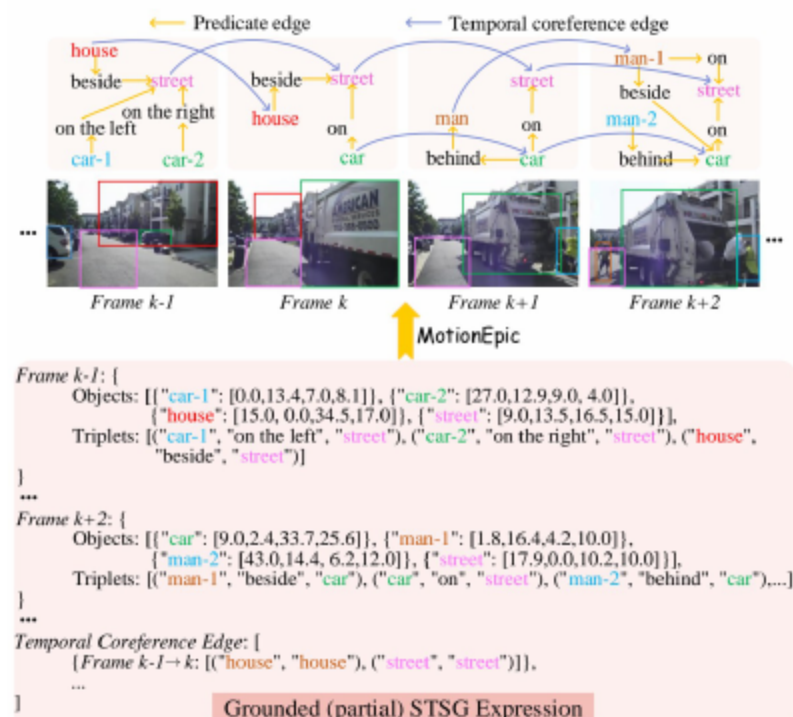
05 MotionEpic Overview and Inputs



- Inputs: text, video, and STSG
- Backbone: Vicuna-7B; Video: ViT-L/14 plus Q-Former; STSG: recurrent Graph Transformer
- Aligned multi-source encodings for grounding-aware reasoning
- **Design**: structured nodes and relations augment dense visual tokens
- **Effect**: improved alignment between content and semantics

[Chiang et al., 2023; Dosovitskiy et al., 2020; Li et al., 2023a]
[Dwivedi & Bresson, 2020]

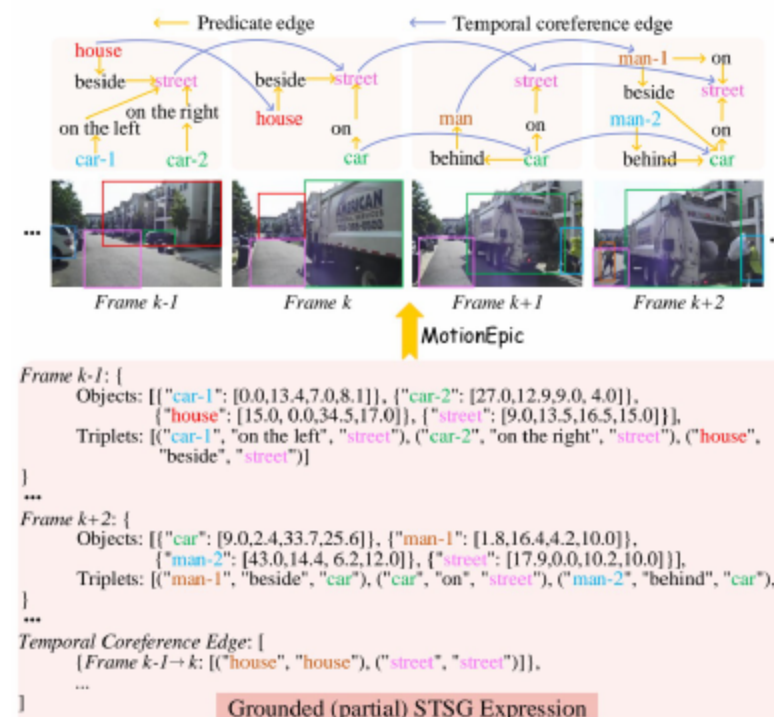
06 Representation and Encoding Design



- STSG: per-frame scene graphs plus temporal coreference edges
- Nodes carry category, features, and bounding box; edges capture predicates and tracking
- Sampling and recurrent propagation improve efficiency and temporal continuity
- **Goal**: robust trajectories and relations for long-range reasoning
- **Risk**: over-sampling adds redundancy and noise; balanced rate preferred

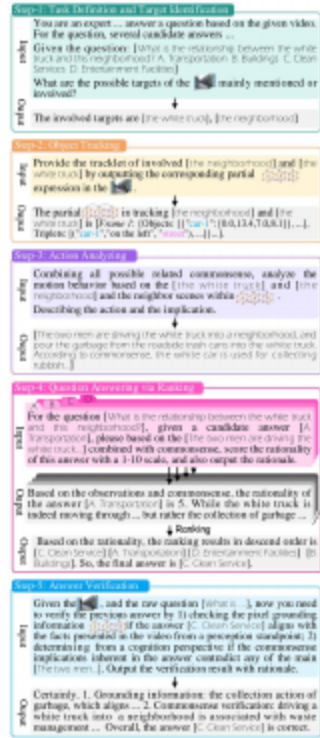
[Ji et al., 2020]

07 Grounded Generation and STSG-free Inference



- Model can generate partial STSG as rationale
- Grounding-aware tuning enables STSG-free downstream inference
- Coarse-to-fine objectives align video and STSG for accurate parsing
- **Recycle**: generated structure feeds subsequent reasoning rounds
- **Outcome**: factual grounding reduces hallucination in answers

08 Video-of-Thought: Perception-to-Cognition Chain



- Steps: target select → tracklet grounding → interaction analysis → candidate ranking → dual verification
- Verification: pixel grounding plus commonsense
- Decomposition yields factual, explainable answers for complex VideoQA
- **Key idea:** solve low-level perception before high-level reasoning
- **Impact:** stronger robustness on long and cluttered videos

[Wei et al., 2022]

10 Dataset-specific: Causal-VidQA and NExT-QA

Model	Acc@D	Acc@E	Acc@P			Acc@C		
			A	R	AR	A	R	AR
● SoTA baselines								
TranSTR	73.6	75.8	65.1	65.0	48.9	68.6	65.3	50.3
Video-LLaMA	69.2	71.0	63.6	62.4	44.4	65.4	60.1	45.0
VideoChat	72.9	73.9	65.2	63.1	45.9	66.0	62.7	45.8
Video-ChatGPT	73.1	75.1	66.0	63.9	46.0	67.8	63.6	50.0
Video-LLaVA	73.7	74.4	67.6	65.4	47.7	68.0	64.9	51.5
● CoT								
Video-LLaVA	74.2	74.8	68.0	65.7	48.1	70.3	65.7	52.9
Video-LLaVA+STSG	75.7	75.9	68.9	67.2	50.0	70.7	67.2	53.6
MotionEpic	<u>78.5</u>	<u>77.2</u>	<u>70.1</u>	<u>70.8</u>	<u>52.4</u>	<u>71.2</u>	<u>69.1</u>	<u>55.0</u>
● VoT								
MotionEpic	81.2	83.0	74.3	73.7	54.7	74.5	73.8	58.6

- Large margins on causal and complex reasoning subsets
- Benefits in description, explanation, prediction, and counterfactual
- Consistent gains across Acc@All, causal, temporal, and descriptive splits
- **Stronger:** reasoning about interactions and intentions
- Stable improvements across multiple baselines and settings

09 Main Results Overview and Key Observations

Model	VLEP	STAR				IntentQA	Social-IQ	
		Int.	Seq.	Pre.	Fea.		2-Way	4-Way
● SoTA baselines								
InternVideo	63.9	62.7	65.6	54.9	51.9	-	-	-
LLaMA-VQA	<u>71.0</u>	66.2	67.9	57.2	52.7	-	-	-
VLAP	69.6	<u>70.0</u>	<u>70.4</u>	<u>65.9</u>	<u>62.2</u>	-	-	-
SeViLA	68.9	63.7	70.4	63.1	62.4	-	-	-
VideoChat	62.0	63.2	66.8	54.1	49.6	59.3	67.7	37.8
Video-LLaVA	65.8	64.3	67.0	56.5	50.1	62.5	68.9	39.2
● CoT								
Video-LLaVA	65.7	65.0	67.7	57.8	52.0	63.2	69.5	40.4
Video-LLaVA+STSG	67.0	65.9	68.9	58.7	53.7	64.9	70.4	41.7
MotionEpic	68.2	66.8	69.6	60.6	57.4	<u>66.1</u>	<u>71.7</u>	<u>43.0</u>
● VoT								
MotionEpic	73.4	71.5	72.6	66.6	62.7	70.8	72.8	45.0

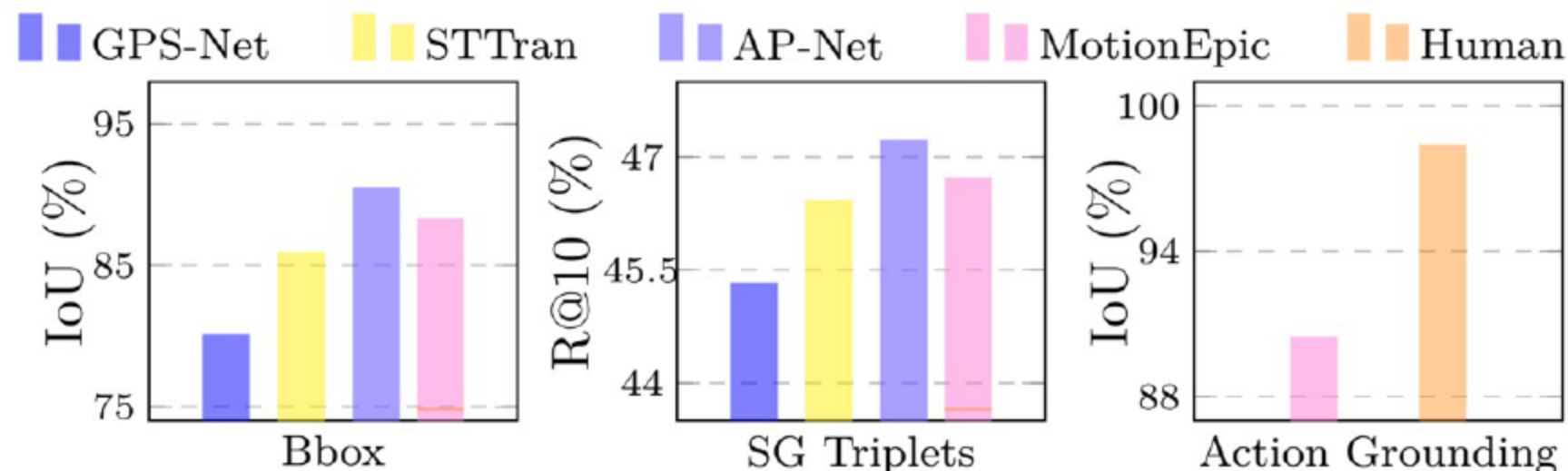
- **Overall:** VoT plus MotionEpic outperforms state of the art across complex VideoQA
- STSG integration and VoT surpass vanilla CoT approaches
- Implicit STSG integration in MotionEpic beats explicit feature fusion
- **Consistent gains:** fine-tuning and zero-shot settings
- CoT alone shows limited lift for video reasoning compared to VoT

11 Zero-shot Performance and Verification Ablations

Model	MSR-VTT	ActivityNet	NExT-QA	STAR	AVG.
• Zero-shot SoTA baselines					
InternVideo	-	-	49.1	41.6	-
Video-LLaMA	49.6	21.4	43.5	36.4	37.7
VideoChat	52.0	26.5	52.8	45.0	44.1
Video-ChatGPT	54.3	35.2	53.0	48.7	47.8
Video-LLaVA	59.2	45.3	57.3	50.6	53.1
VideoChat2	54.1	49.1	61.7	59.0	56.0
• CoT					
Video-LLaVA	60.0	46.9	59.5	52.0	54.6
Video-LLaVA+STSG	61.5	48.4	60.6	52.7	55.8
MotionEpic	63.1	50.0	61.9	56.5	57.8
• VoT					
MotionEpic	66.2	54.6	66.5	61.7	62.3
w/o Verify-G	63.6	51.4	62.0	59.1	59.0
w/o Verify-C	65.1	53.4	62.8	58.8	60.1

- VoT beats CoT in zero-shot; verification boosts robustness
- Grounding verification critical for perceptive datasets; both matter for complex videos
- Dual verification yields clearest gains on complex QA (NExT-QA, STAR)
- **Finding:** VoT benefits grow when no in-domain training is used
- **Net effect:** higher accuracy through evidence-backed answers

12 Grounding Probes and Ablations



- MotionEpic competitive with STSG parsers on grounding tasks
- Ablations reveal which grounding objectives matter most
- Reliable grounding supports downstream multi-

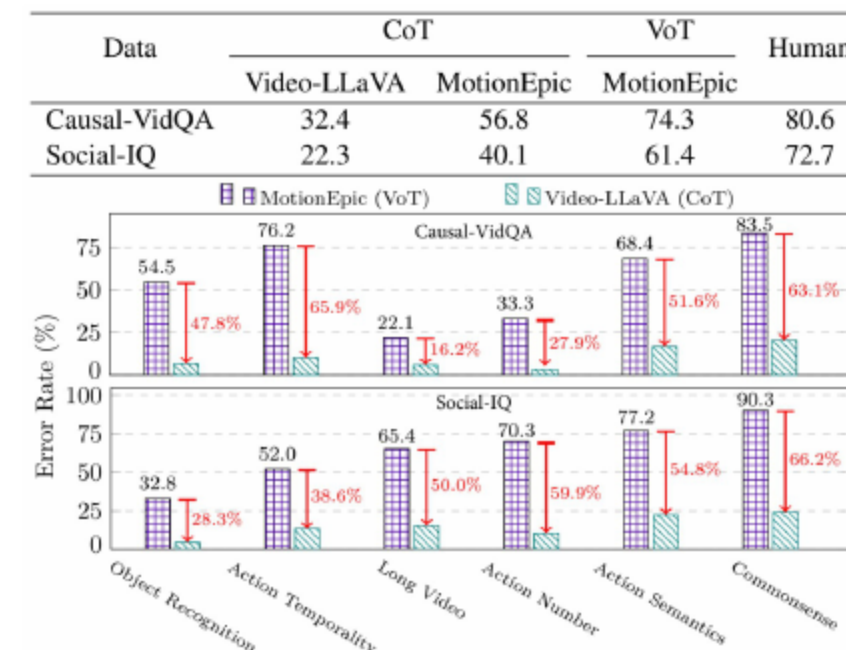
step reasoning

- **Evidence:** strong object, relation, and action alignment
- **Takeaway:** grounding quality correlates with reasoning accuracy

[Lin et al., 2020; Cong et al., 2021; Li et al., 2022b]

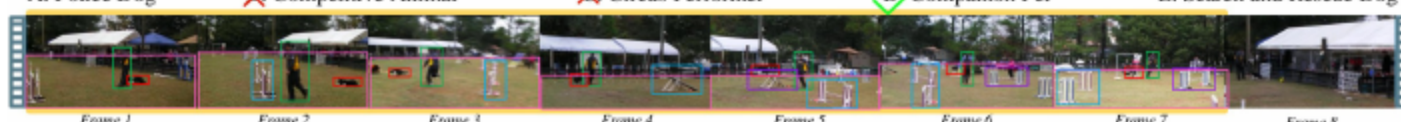
13 Error Breakdown and Human Eval

- MotionEpic plus VoT approaches human performance on hard subsets
- Error reductions in action semantics and commonsense
- Human study validates VoT gains over CoT baselines
- **Improved:** interpretability and answer faithfulness
- **Focus:** fewer mistakes in multi-entity interactions



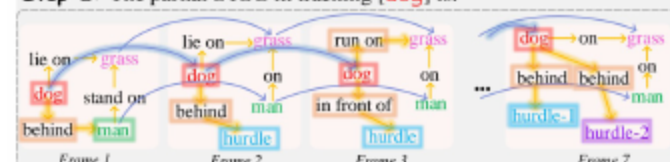
14 End-to-end VoT Case

Question: What is the least likely category for the animal in this video?
A. Police Dog ☒ Competitive Animal ☒ Video-LLaVA ☒ Circus Performer ☒ Video-ChatGPT ☒ D. Companion Pet ☒ MotionEpic ☒ E. Search and Rescue Dog



Step-1: The involved target is [dog].

Step-2: The partial STSG in tracking [dog] is:



Step-3: According to the video scene and STSG, the dog is **crossing multiple hurdles** with the dog being visible both before and after the hurdles. The **accompanying** man is observed providing instructions to guide the dog through the obstacles... **Drawing on factual commonsense understanding**, it might be inferred that the man is a trainer who is **imparting various commands and training the dog on a grassy field**.

Step-4:

The video depicting professional training and complex actions suggests it might be a police dog performing daily training ... The rationality of the answer [A. Police Dog] is 2.

The companion dog is to support companionship and emotional support to their owners rather than engaging in specialized tasks ... The answer [D. Companion Pet] has a coherence score of 8.

After ranking the rationale score, the final answer is [D. Companion Pet].

Step-5: Let's verify the [D. Companion Pet] based on visual perception ..

1. Pixel Grounding Information Check: Based on the video scene, it depicts a training ground with a dog, so the answer is fitting.
2. Commonsense Check: Observing the dog's energetic behavior during training aligns with the common understanding that companion pet are less likely to undergo such training, supporting the chose answer.
Conclusion: The answer [D. Companion Pet] is supported both by ...

- STSG grounding reduces hallucination
- Sequential reasoning and verification yields correct answer
- Explainable chain from perception to cognition

- **Rationale:** grounded entities, relations, and actions
- **Clarity:** traceable steps support human auditing

15 VoT + MotionEpic—From Grounding to Cognition



- **VoT:** first video-native CoT framework with dual verification
- **MotionEpic:** STSG-grounded video MLLM enabling STSG-free inference
- State-of-the-art across complex VideoQA; broad implications
- **Outlook:** transferable to richer multimodal reasoning tasks
- Bridging perception and cognition for reliable video understanding

An abstract geometric design featuring a central white parallelogram with a teal border and a drop shadow, containing the text 'THANKS!'. This central element is flanked by teal rectangular blocks. The composition is decorated with various teal geometric shapes, including triangles and lines, scattered around the central area. A thin black horizontal line runs across the middle of the image, and a thin black vertical line runs down the right side, intersecting at the center.

THANKS!